

Detection in Crowded Scenes: One Proposal, Multiple Predictions

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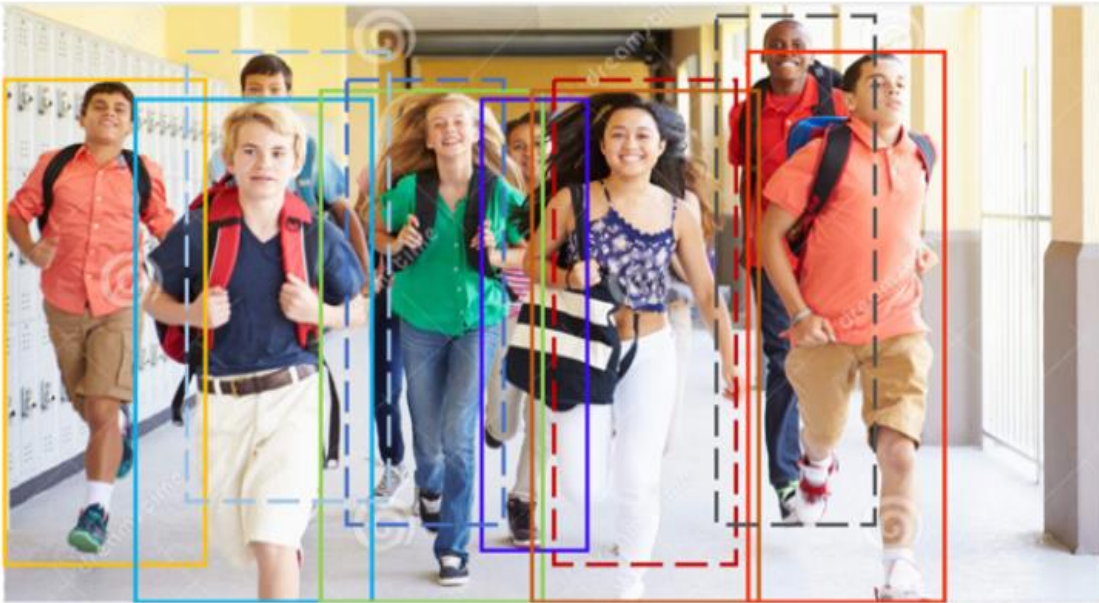
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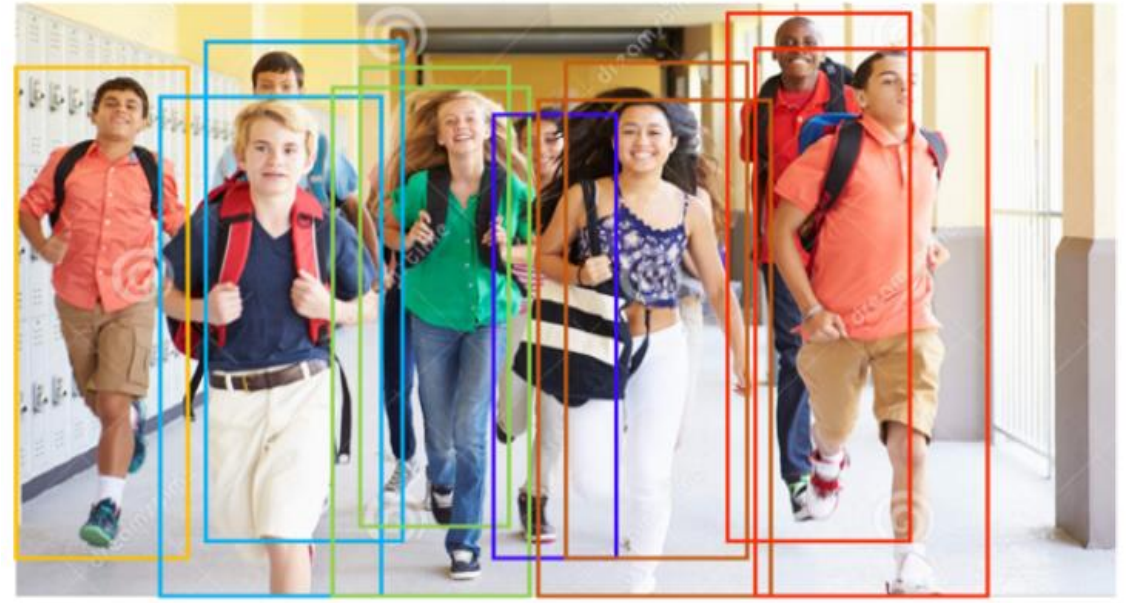
목차

1. Introduction
2. Proposed Approach: Multiple Instance Prediction
3. Experiment
4. Conclusion & Discussion

Introduction – Crowded Object Detection



(a) Baseline



(b) Ours

Contribution

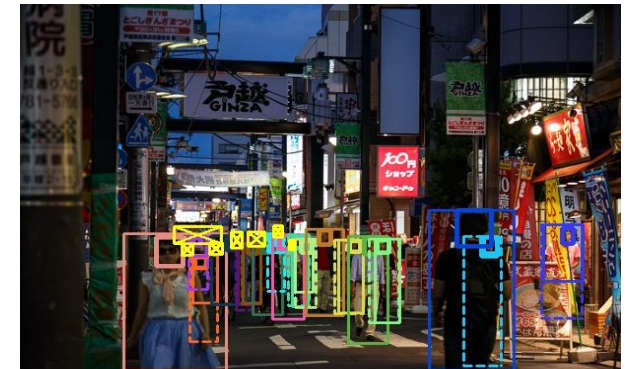
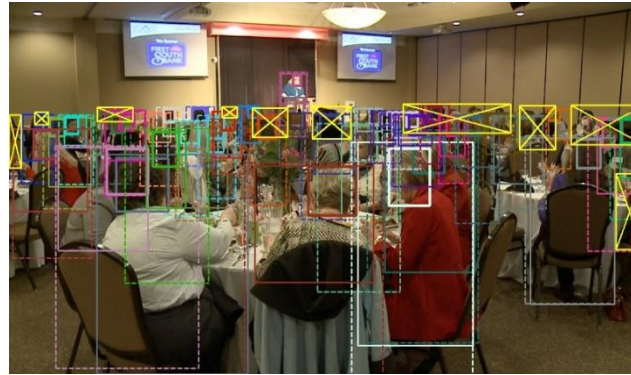
1. Proposed a novel approach: Multiple Instance Prediction
2. Proposed a novel loss: EMD loss
3. Proposed a novel NMS: Set NMS
4. Achieved SOTA on CrowdHuman Dataset

CrowdHuman Dataset

- train/val/test: 15,000 / 4,370 / 5,000
- 470K human instances

Dataset	# objects/img	# overlaps/img
CrowdHuman [32]	22.64	2.40
CityPersons [42]	6.47	0.32
COCO* [22]	9.34	0.015

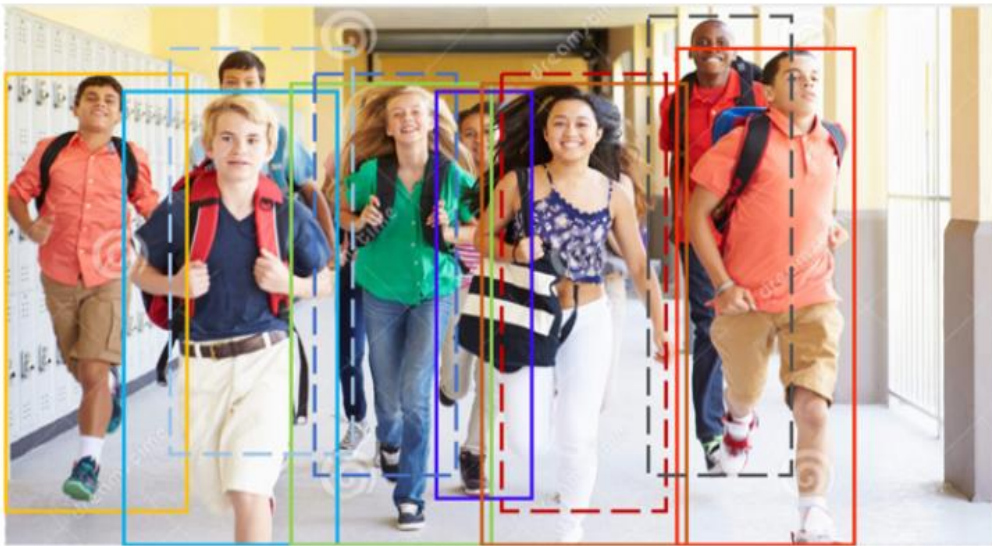
Table 1. *Instance density* of each dataset. The threshold for overlap statistics is $\text{IoU} > 0.5$. *Averaged by the number of classes.



- Shao et al., 2018, CrowdHuman: A Benchmark for Detection Human in a Crowd
- <https://www.crowdhuman.org/>

Fundamental difficulties in crowded object detection

- State-of-the-art models on COCO or VOC perform poorly on CrowdHuman dataset
 - Highly overlapped instances are likely to have very similar features
 - Heavily overlapped instances are likely to be mistakenly suppressed by NMS



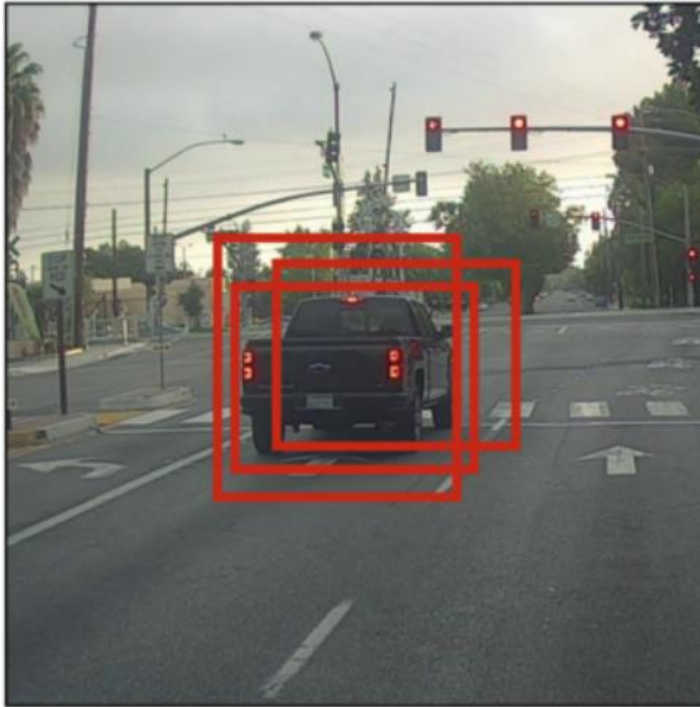
(a) Baseline

Algorithm 1 Non-Max Suppression

```
1: procedure NMS( $B, c$ )
2:    $B_{nms} \leftarrow \emptyset$ 
3:   for  $b_i \in B$  do
4:      $discard \leftarrow \text{False}$ 
5:     for  $b_j \in B$  do
6:       if  $\text{same}(b_i, b_j) > \lambda_{nms}$  then
7:         if  $\text{score}(c, b_j) > \text{score}(c, b_i)$  then
8:            $discard \leftarrow \text{True}$ 
9:       if not  $discard$  then
10:         $B_{nms} \leftarrow B_{nms} \cup b_i$ 
11:   return  $B_{nms}$ 
```

NMS (Non-Maximum Suppression)

Before non-max suppression



Non-Max
Suppression

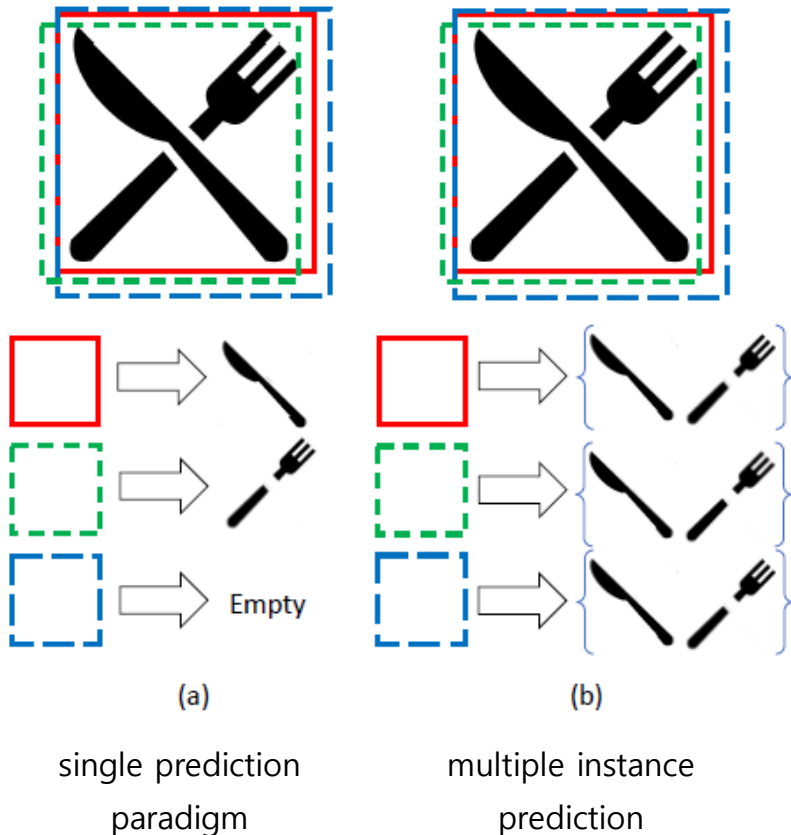


After non-max suppression



Solution – multiple instance prediction

- For each proposal box, rather than predicting a single instance, propose a **set of instances**

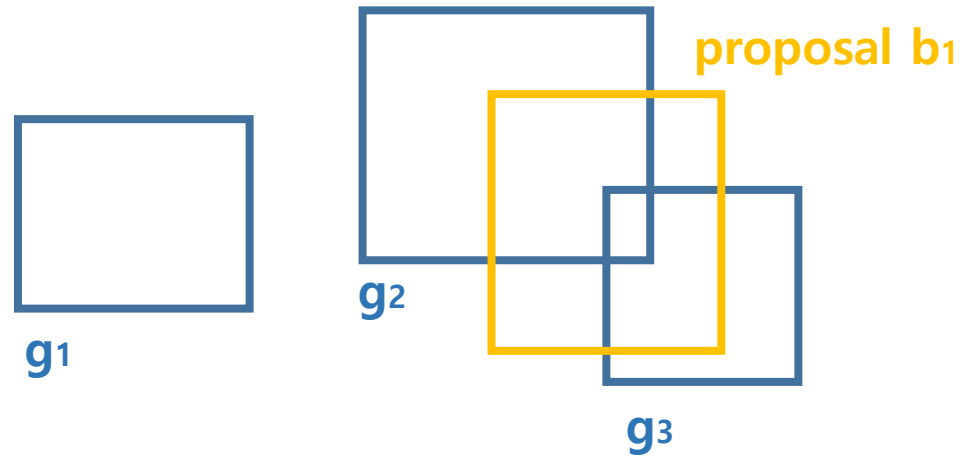


- (a) Each proposal box predicts a single instance (intrinsically difficult!). After NMS, only **one prediction survives**.
- (b) Set NMS removes duplicates from different proposals while **keeping duplicates** in a proposal.

Solution – multiple instance prediction

- Step 1: assign a proposal box to ground-truths

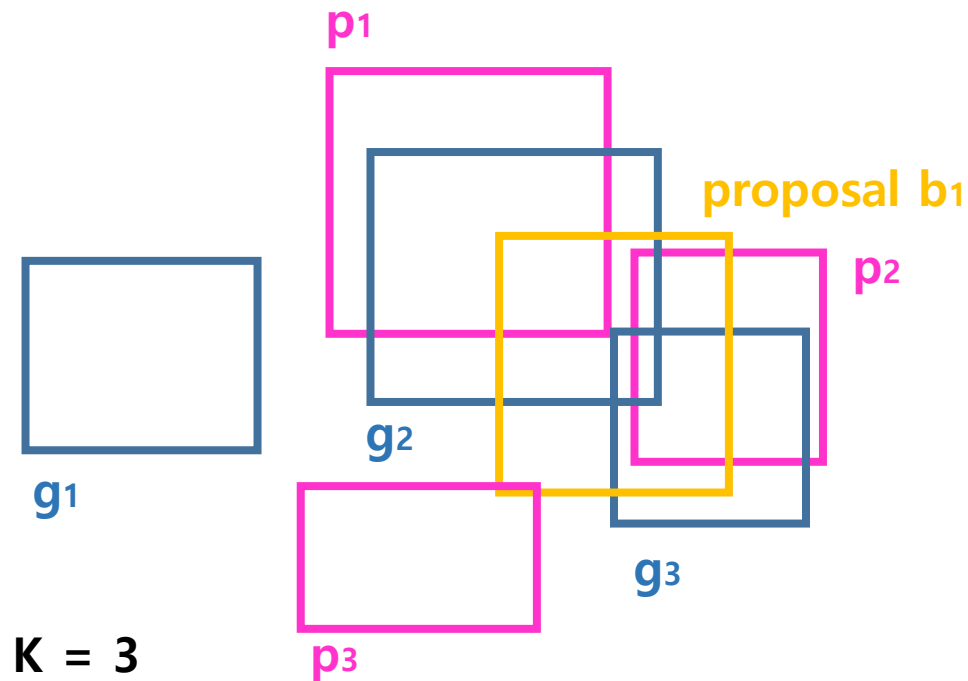
$$G(b_i) = \{g_j \in \mathcal{G} | \text{IoU}(b_i, g_j) \geq \theta\}, \quad (1)$$



Solution – multiple instance prediction

- Step 2: make K predictions from one proposal box

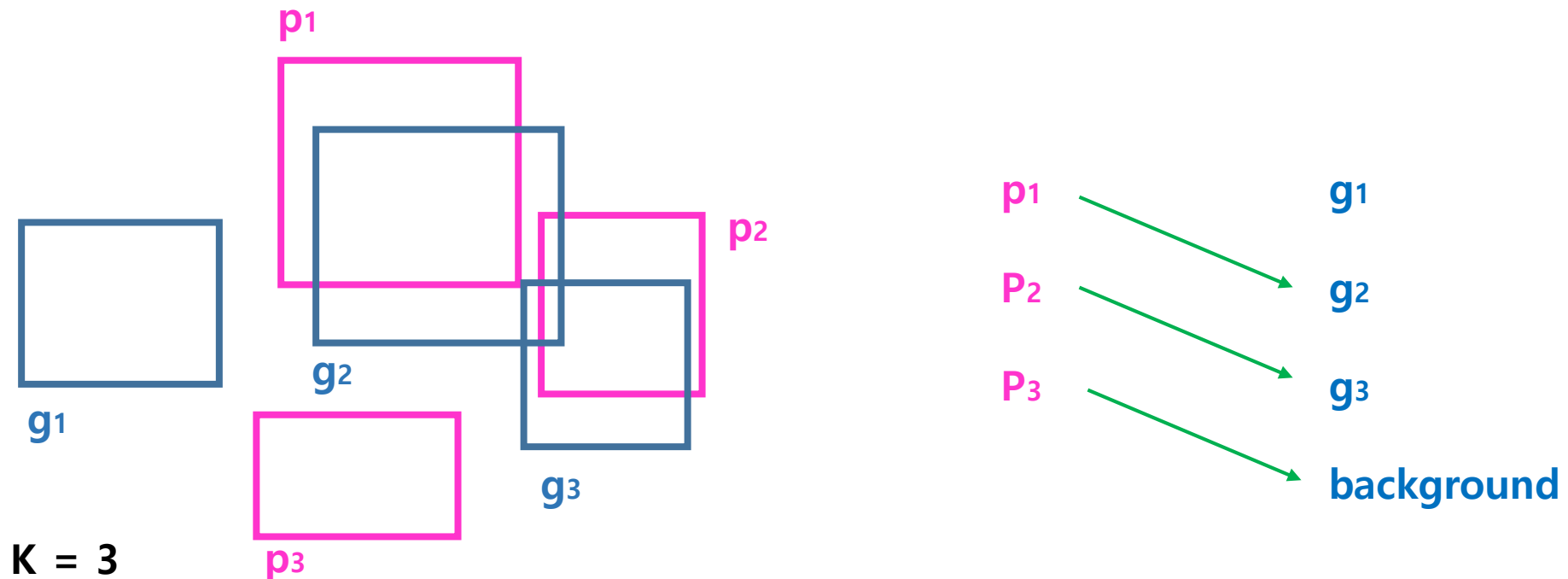
$$P(b_i) = \left\{ (\mathbf{c}_i^{(1)}, l_i^{(1)}), (\mathbf{c}_i^{(2)}, l_i^{(2)}), \dots, (\mathbf{c}_i^{(K)}, l_i^{(K)}) \right\}, \quad (2)$$



EMD Loss

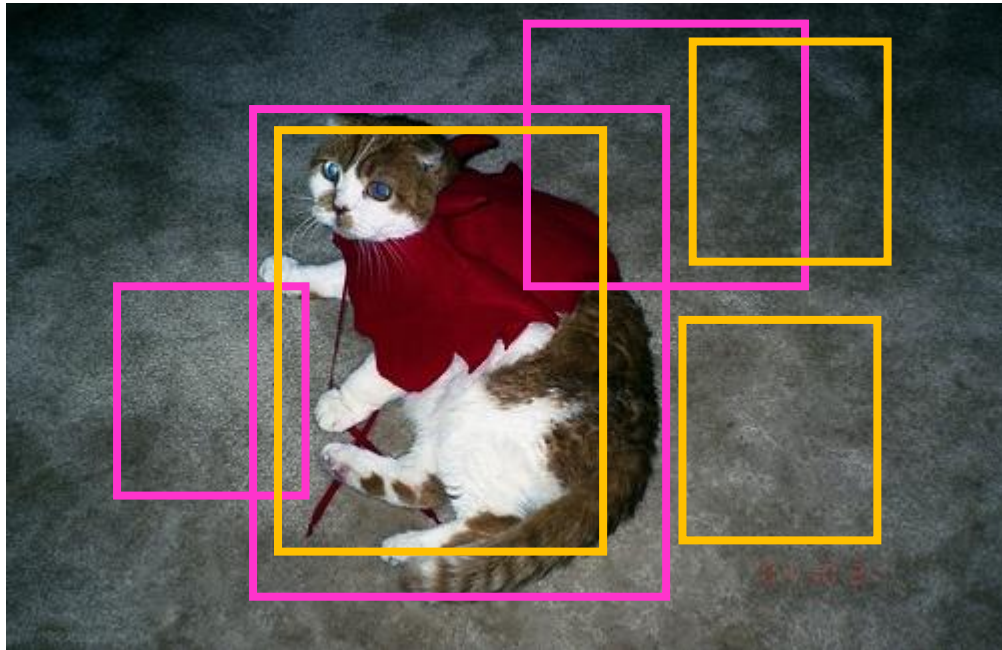
- Step 3: assign predictions to ground-truths using Earth Mover's Distance (EMD)

$$\text{EMD loss: } \mathcal{L}(b_i) = \min_{\pi \in \Pi} \sum_{k=1}^K \left[\mathcal{L}_{cls}(\mathbf{c}_i^{(k)}, g_{\pi_k}) + \mathcal{L}_{reg}(\mathbf{l}_i^{(k)}, g_{\pi_k}) \right] \quad (3)$$



Set NMS

- Step 4: apply Set NMS



Set NMS



Set NMS



Architecture

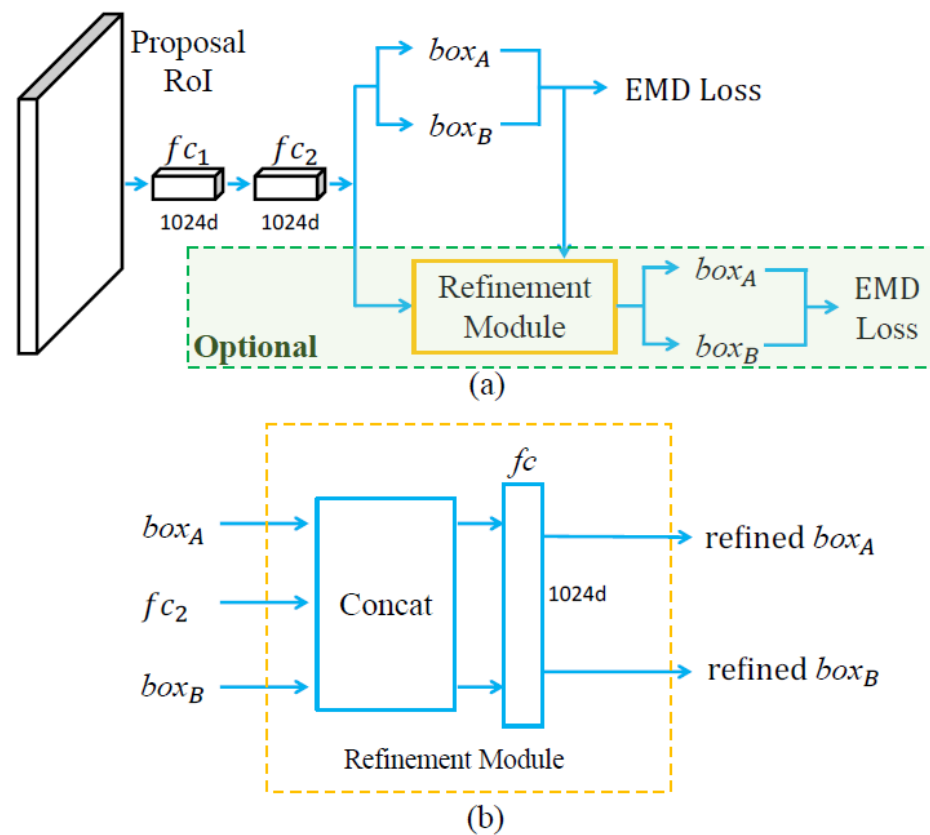


Figure 3. Overall architecture. (a) box_A and box_B are the two instances predicted by one proposal, using our EMD Loss. *Refinement module* is an optional step. (b) The refinement module concatenates feature and box information to optimize the results.

Q & A

Experiments

- **Evaluation Metrics**

- 1) Averaged Precision (AP)
- 2) MR^{-2} Miss Rate on False Positive Per Image (FPPI) in $[10^{-2}, 100]$
- 3) Jaccard Index

- **Datasets**

- 1) CrowdHuman
- 2) CityPersons
- 3) COCO

Dataset	# objects/img	# overlaps/img
CrowdHuman [32]	22.64	2.40
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Table 1. *Instance density* of each dataset. The threshold for overlap statistics is $IoU > 0.5$. *Averaged by the number of classes.

- **Network Architecture**

- 1) Backbone: ResNet-50 pre-trained on ImageNet
- 2) Head: FPN with RoIAlign
- 3) $K = 2$

Main results and ablation study

Performance on CrowdHuman Dataset

MIP	Set NMS	RM	AP/%	MR ⁻² /%	JJ/%
			85.8	42.9	79.8
✓			87.4	42.8	80.8
✓	✓		90.3	42.2	82.0
✓	✓	✓	90.7	41.4	82.3

Table 2. Ablation experiments evaluated on *CrowdHuman* validation set. The baseline model (the first line) is our reimplemented *FPN* [20] with *ResNet-50* [15] backbone. *MIP* – multiple instance prediction with *EMD loss*. *RM* – refinement module.

Comparison with various NMS strategies

Performance on CrowdHuman Dataset

Method	IoU*	AP/%	$MR^{-2}/\%$	J1/%
NMS	0.3	72.3	48.5	69.6
	0.4	80.7	44.6	76.3
	0.5	85.8	<u>42.9</u>	<u>79.8</u>
	0.6	<u>88.1</u>	45.4	79.4
	0.7	87.1	56.5	74.4
	0.8	82.8	68.5	62.6
Soft-NMS [1]	0.5	88.2	42.9	79.8
Ours (w/o RM)	0.5	90.3	42.2	82.0
Ours (with RM)	0.5	90.7	41.4	82.3

Table 3. Comparisons of different NMS strategies on *CrowdHuman* validation set. The baseline model is *FPN* [20]. *IoU threshold for post-processing. *RM* – refinement module.

Ablation on Number of Heads

Performance on CrowdHuman Dataset

	AP/%	MR ⁻² /%	JI/%
$K = 1$	85.8	42.9	79.8
$K = 2$	90.7	41.4	82.3
$K = 3$	90.7	41.6	82.1

Table 10. Ablation experiments evaluated on the *CrowdHuman* validation set. It worth noting that if $K = 1$, the architecture is the same as the *single-instance-prediction* baseline.

Method	AP/%	MR ⁻² /%	JI/%
RetinaNet Baseline	82.0	56.3	73.2
Ours	82.7	54.7	74.0

Table 11. Experiment on RetinaNet[21]. All hyperparameters between the RetinaNet baseline and our method remain the same. The results show that one-stage detectors can also benefit from our method.

Experiments on COCO

Method	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
FPN	37.5	59.6	40.4	23.0	41.2	48.6
Soft-NMS	38.0	59.4	41.5	23.3	41.8	49.0
Ours	38.5	60.5	41.5	23.0	41.8	50.1

Table 8. Comparisons on *COCO* [22] *minival* set. All models are based on *FPN* detector. Results are evaluated in all the 80 classes.

Conclusion & Discussion

1. Proposed approach is not only effective on crowded scenes, but also generalizes well on normal data.
2. Proposed approach is compatible with other one-stage & two-stage architectures.
3. A local version of DETR (Carion et al., 2020)?

Thank you